$$A = \begin{pmatrix} 3 & 4 & 7 \\ 2 & 1 & 3 \\ 7 & 2 & 1 \end{pmatrix}$$

**Solution:**

Determinant of the given matrix is,

$$|A| = \begin{vmatrix} 3 & 4 & 7 \\ 2 & 1 & 3 \\ 7 & 2 & 1 \end{vmatrix} = 3\begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix} - 4\begin{vmatrix} 2 & 3 \\ 7 & 1 \end{vmatrix} + 7\begin{vmatrix} 2 & 1 \\ 7 & 2 \end{vmatrix}$$

$$= 3\,(1-6) - 4\,(2-21) + 7\,(4-7)$$

$$= 3\,(-5) - 4\,(-19) + 7\,(-3)$$

$$= -15 + 76 - 21$$

$$\boxed{|A| = 40}$$

The value of determinant is 40.

## 12.3.3 Cramer's Rule



G. CRAMER
(1704-1752)

Cramer's rule provides the solution of a system of linear equations with '**n**' variables and '**n**' equations. It helps to arrive at a unique solution of a system of linear equations with as many equations as unknowns.

If the given equations are

$$a_{11}x + a_{12}y + a_{13}z = b_1$$

$$a_{21}x + a_{22}y + a_{23}z = b_2$$

$$a_{31}x + a_{32}y + a_{33}z = b_3$$

then

$$x = \frac{\Delta x}{\Delta},\quad y = \frac{\Delta y}{\Delta},\quad z = \frac{\Delta z}{\Delta}$$

where, $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$

$$\Delta_x = \begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix}, \Delta_y = \begin{vmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{vmatrix},$$

$$\Delta z = \begin{vmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{vmatrix}$$

**Key Note**

If the determinant $\Delta = 0$, then solution does not exist.

**Example: 12.8**

Find the value of x and y in the equations by using Cramer's rule. $x + 3y = 1$ and $3x - 2y = 14$

**Solution:**

Given equations are

$$x + 3y = 1$$
$$3x - 2y = 14$$

Then the equations in the matrix form :

$$\begin{pmatrix} 1 & 3 \\ 3 & -2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 14 \end{pmatrix}$$

Calculating $\Delta$, $\Delta = \begin{vmatrix} 1 & 3 \\ 3 & -2 \end{vmatrix}$

$$= -2 - 9$$
$$= -11$$

$\Delta \neq 0$, Hence solution exists.

$$\Delta_x = \begin{vmatrix} 1 & 3 \\ 14 & -2 \end{vmatrix} = -2 - 42 = -44$$

$$\Delta_y = \begin{vmatrix} 1 & 1 \\ 3 & 14 \end{vmatrix} = 14 - 3 = 11$$



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